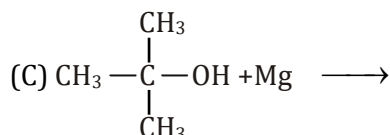
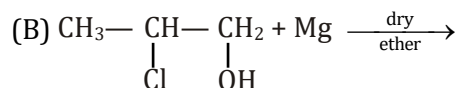
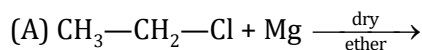


EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

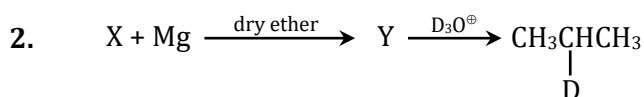
Preparation of Grignard reagent & Acid Base Reaction

1. Grignard reagent can be prepared by

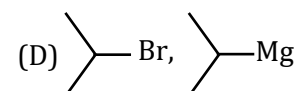
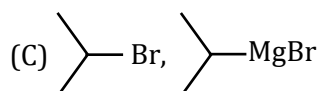
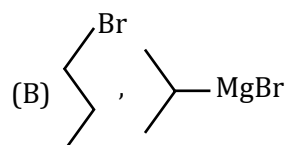
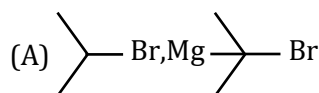


(D) All of them

CGR060



Identify X and Y –



CGR061

3. The correct order of reactivity of halides with Mg for the formation of Grignard reagents :-

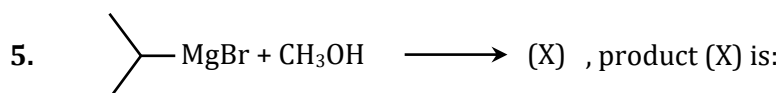


CGR062

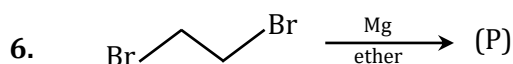
4. Propane is not formed when
- $\text{C}_3\text{H}_7\text{MgBr}$
- is treated with :-



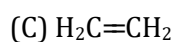
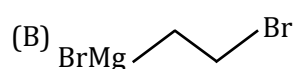
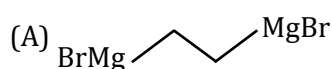
CGR063



CGR064



(P) is :-



(D) All

CGR065

7. $C_2H_5MgBr + CH_3-NH_2 \longrightarrow (P)$
Product (P) is :-
(A) C_2H_6 (B) C_3H_8 (C) $C_2H_5NH_2$ (D) No reaction

CGR066

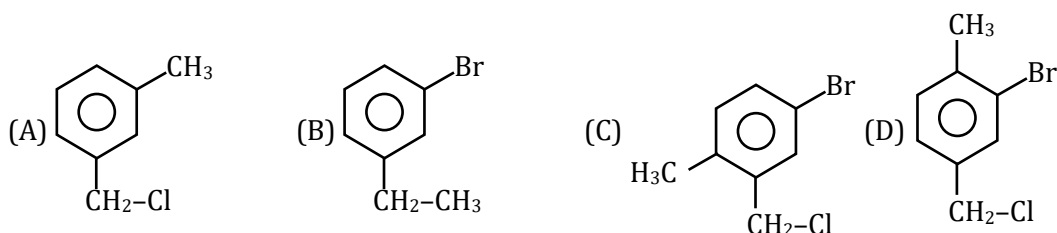
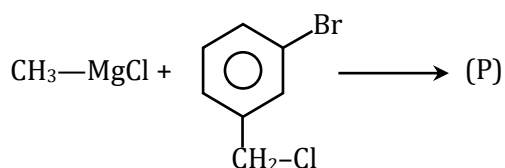
8. Which gas is released during the following reaction :-
 $HC\equiv CH + MeMgBr \longrightarrow$
(A) Ethane (B) Propane (C) Methane (D) Butane

CGR067

9. Which of the following will undergo acid-base reaction with Grignard reagent :-
(A) $HC\equiv CH$ (B) $R-OH$ (C) $R-CO_2H$ (D) All of these

CGR068

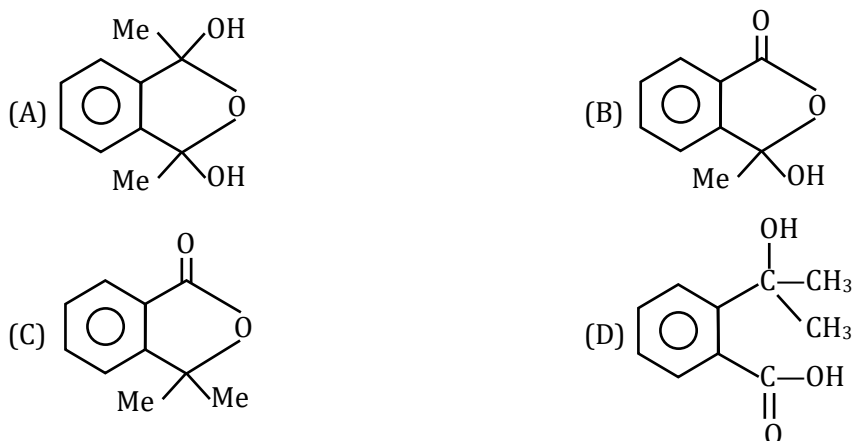
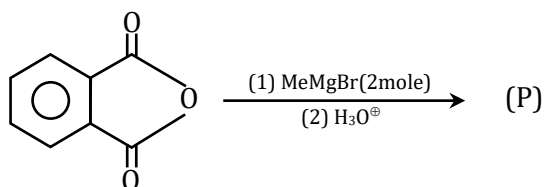
10. Find the major product for the given following reaction :



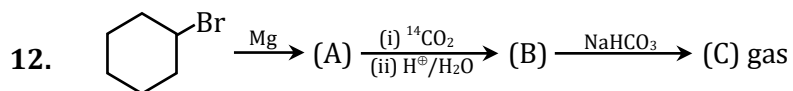
CGR069

Chemical Reaction of Grignard Reagents (NAER, NAR, SN Reaction)

11. For the reaction the product expected is :



CGR070



Product C is

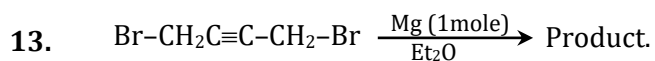
(A) CO

(B) $^{14}\text{CO}_2$

(C) CO_2

(D) A mixture of $^{14}\text{CO}_2$ and CO_2

CGR073



The major product is ?

(A) $\text{Br}-\text{Mg}-\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_2-\text{Br}$

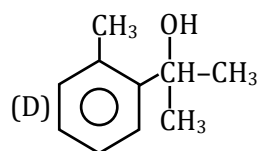
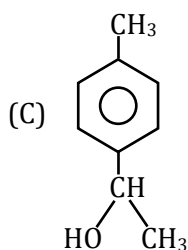
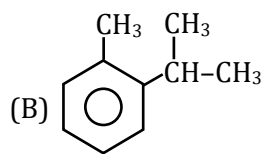
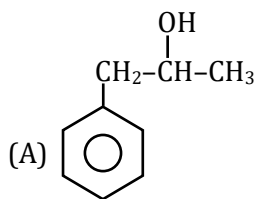
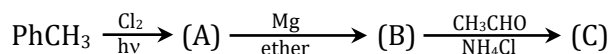
(B) Cyclobutyne

(C) $-\{\text{CH}_2-\text{C}\equiv\text{C}-\text{CH}_2\}_n$

(D) $\text{CH}_2=\text{C}=\text{C}=\text{CH}_2$

CGR071

14. Find product C of the following reaction :



CGR078

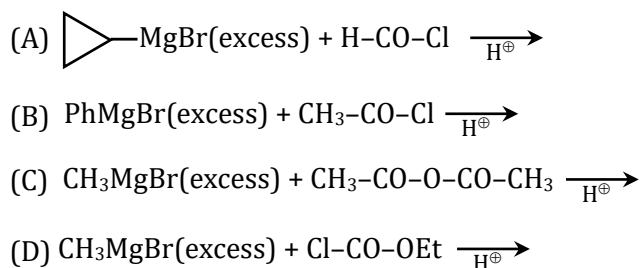
EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Reaction of Grignard's Reagent

1. From which compound Alcohol is formed when RMgX followed by H_3O^+ is treated with :
 (A) Ethanoyl chloride (B) O_2 (C) oxirane (D) trimethyl orthoformate
CGR079

2. In which of the following reactions 3° alcohol will be obtained as a product.



CGR080

3. Select the correct statement :

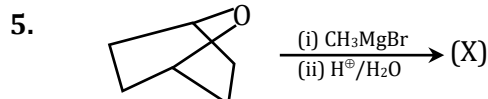
- (A) 1,4-dibromo butane react with excess of magnesium in ether to generate di-Grignard reagent.
 (B) 1,2-dichlorocyclohexane treated with excess of Mg in ether produces cyclohexene.
 (C) Vicinal dihalides undergo dehalogenation to give alkene when heated with Zn dust or Mg.
 (D) 1,3-dichloropropane by treatment with Zn dust or Mg forms cyclopropane.

CGR081

4. Nucleophilic addition of Grignard reagent can not occur in



CGR082



Correct statement about X is

- (A) The product is optically active.
 (B) The product contains plane of symmetry.
 (C) The product shown geometrical isomerism.
 (D) The product shown optical isomerism.

CGR083

EXERCISE (S)

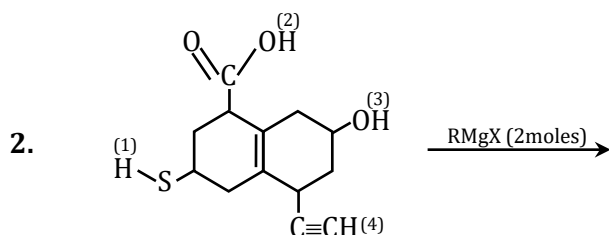
Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Preparation of Grignard reagent & Acid Base Reaction

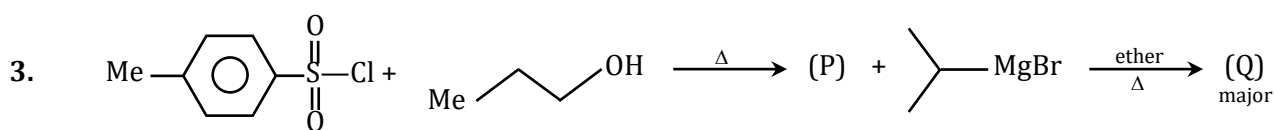
1. How many total possible Grignard reagent can give 3-methylpentane on treatment with H^+ ?

CGR084



The sum of the positions from where deprotonation will occur is

CGR076

Reaction of Grignard's Reagent

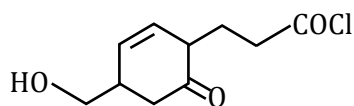
No. of chiral centre in product (Q).

CGR074

4. Compounds are shown with the no. of $RMgX$ required for complete reaction as (x), (y) and (z). what is $x+y+z$?
- (i) $CH_3COOC_2H_5$ (x) (ii) CH_3COCl (y) (iii) $HOCH_2COOC_2H_5$ (z)

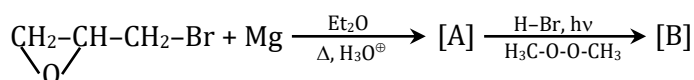
CGR075

5. Number of moles of $RMgX$ required for complete reaction with one mole of the following compound is ?



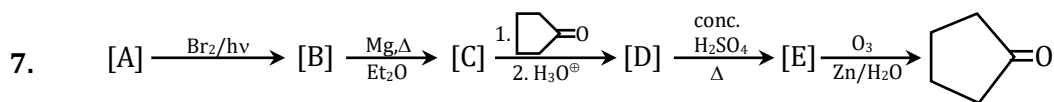
CGR077

6. Choose the final product (B) of the following reaction.



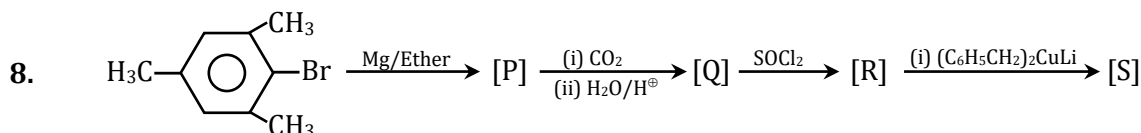
What is the molecular weight of final product ?

CGR127



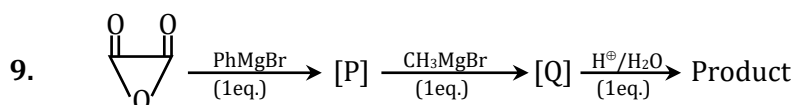
Find D.O.U. in compound [E] ?

CGR128



How many 2° carbon in product [S] ?

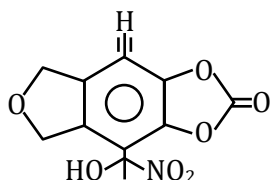
CGR129



Calculate total number of hydrogen atom present in final product ?

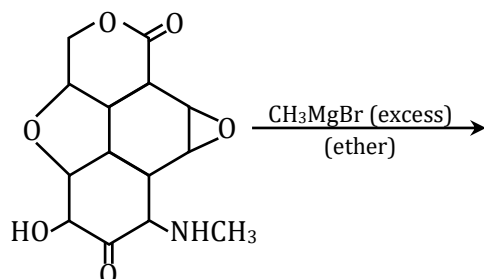
CGR130

10. Calculate the number of molecules of Grignard reagent consumed by 1 molecule of following compound.



CGR089

11. Find the number of CH_3MgBr molecules used the given reactant, to react completely.

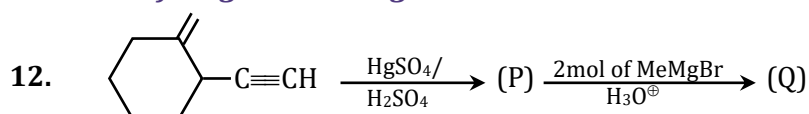


CGR090

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Reaction of Grignard's Reagent

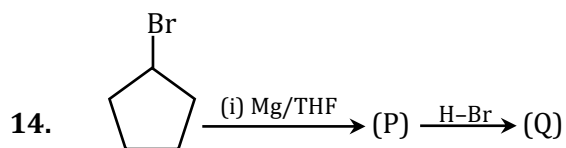


Number of 3° Alcohol formed in product Q

CGR026

13. How many carbonyl isomers of $C_5H_{10}O$ reacts with $PhMgBr$ to give racemic mixture (excluding stereoisomer).

CGR027



No. of sp^3 hybridized carbon in (Q) is :-

CGR028

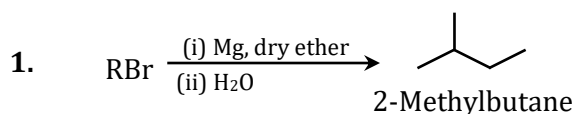
15. How many moles of methane is formed when one mole of lactic acid reacts with excess methyl magnesium chloride ?

CGR029

16. The number of moles of Grignard reagent consumed per mole of the compound is $CH_2(OH)-CH_2-CO-CH_2-CH_2-COEt$

CGR030

EXERCISE – JEE (Main) PYQ



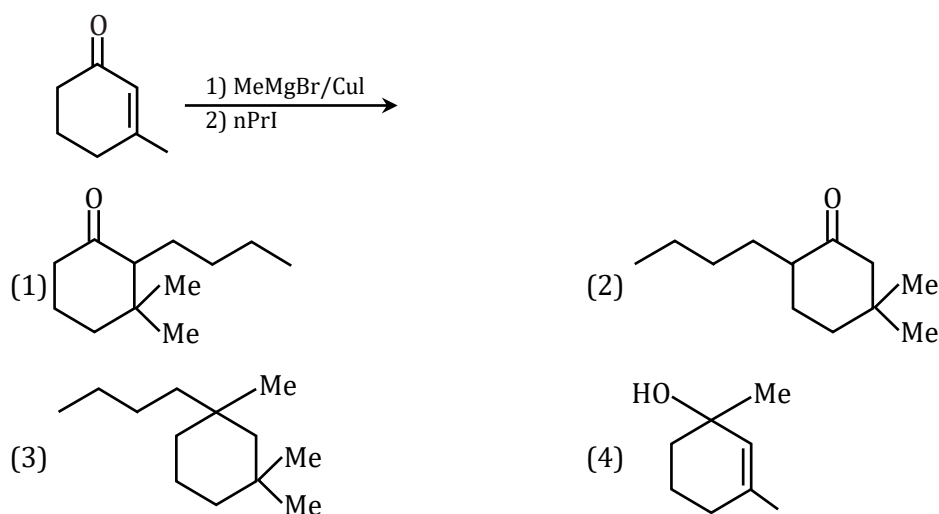
The maximum number of RBr producing 2-methylbutane by above sequence of reactions is ____.
(Consider the structural isomers only) [JEE (Main) 2025]

- (1) 4 (2) 5 (3) 3 (4) 1

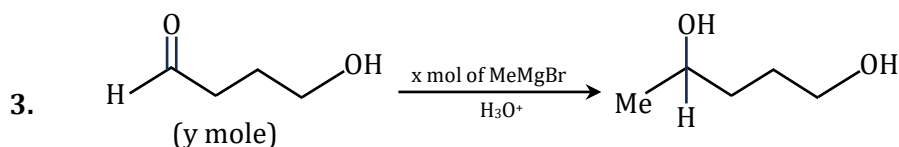
CGR132

2. Find out the major product from the following reaction

[JEE (Main) 2023]



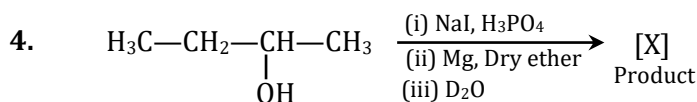
CGR107



The ratio x/y on completion of the above reaction is ____.

[JEE (Main) 2023]

CGR108

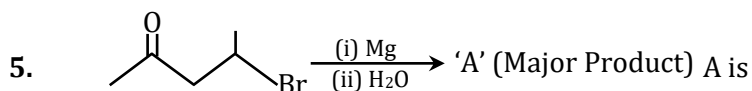


Product [X] formed in the above reaction is :

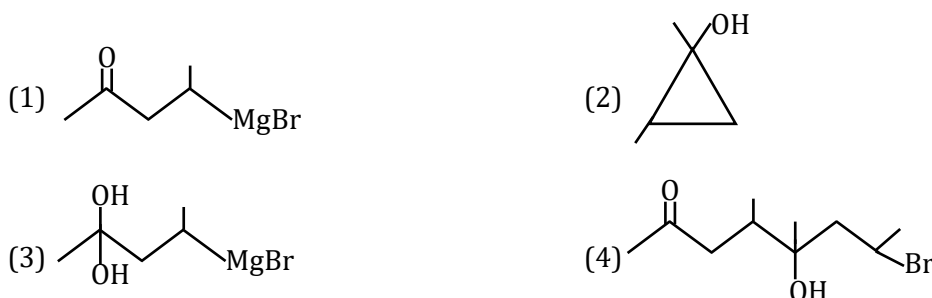
[JEE (Main) 2023]

- (1) $\text{H}_3\text{C}-\text{CH}_2-\underset{\text{D}}{\text{CH}}-\text{CH}_3$ (2) $\text{H}_3\text{C}-\text{CH}_2-\underset{\text{OH}}{\overset{\text{H}}{\text{C}}}-\text{CH}_3$
(3) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}=\text{CH}_2$ (4) $\text{H}_3\text{C}-\text{CH}=\text{CH}-\text{CH}_3$

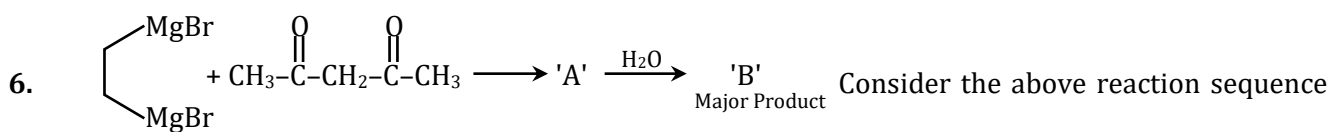
CGR109



[JEE (Main) 2023]

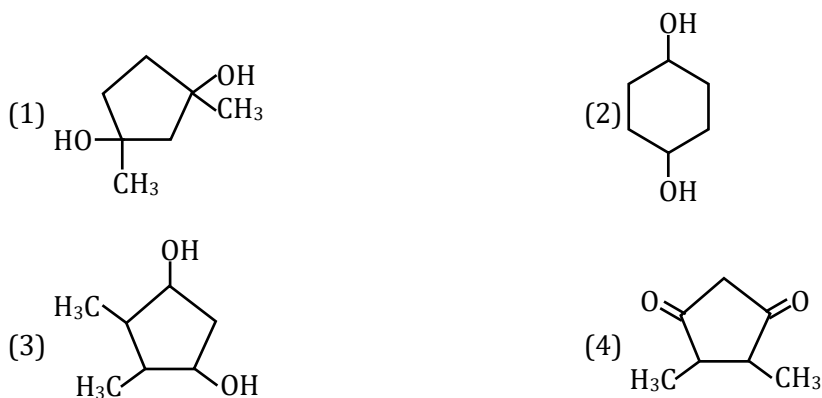


CGR110

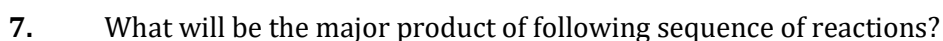


and identify the product B.

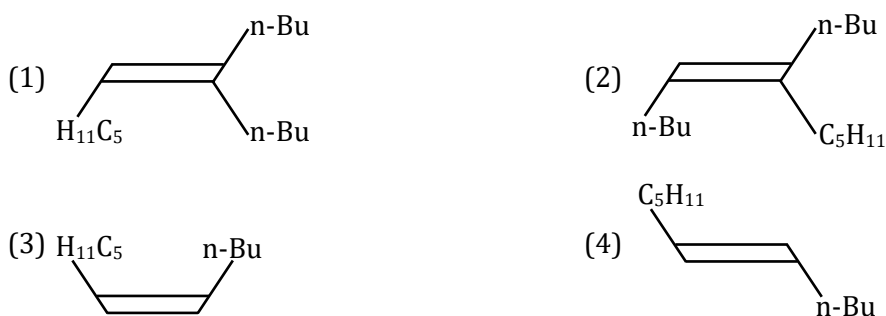
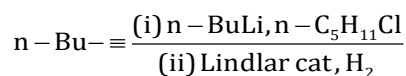
[JEE (Main) 2022]



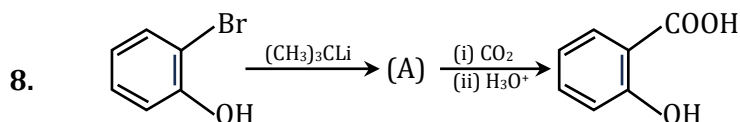
CGR100



[JEE (Main) 2022]

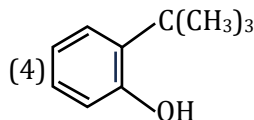
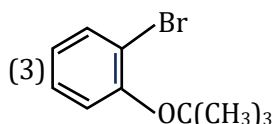
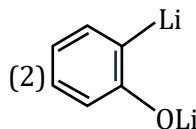
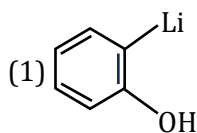


CGR101



In the given conversion the compound A is:

[JEE (Main) 2022]



CGR102

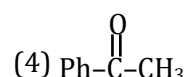
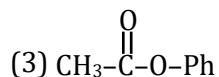
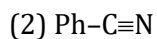
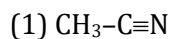
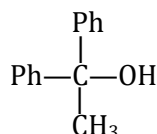
9. A sample of 4.5 mg of an unknown monohydric alcohol, R-OH was added to methylmagnesium iodide. A gas is evolved and is collected and its volume measured to be 3.1 mL. The molecular weight of the unknown alcohol is ____ g/mol. [Nearest integer]

[JEE (Main) 2022]

CGR103

10. Which reactant will give the following alcohol on reaction with one mole of phenyl magnesium bromide (PhMgBr) followed by acidic hydrolysis ?

[JEE (Main) 2022]



CGR104

11. A 100 mL solution of $\text{CH}_3\text{CH}_2\text{MgBr}$ on treatment with methanol produces 2.24 mL of a gas at STP. The weight of gas produced is ____ mg. [nearest integer]

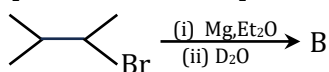
[JEE (Main) 2022]

CGR105

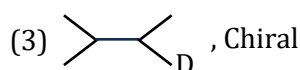
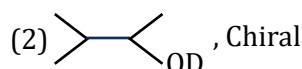
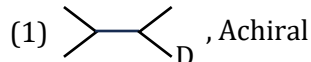
12. Compound 'A' undergoes following sequence of reactions to give compound 'B'. The correct structure and chirality of compound 'B' is:

[JEE (Main) 2022]

[where Et is $-\text{C}_2\text{H}_5$]



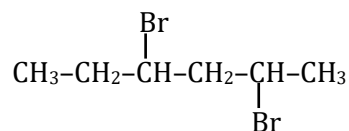
Compound 'A'



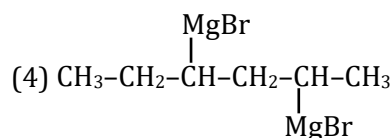
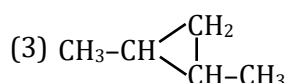
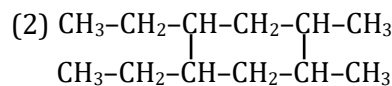
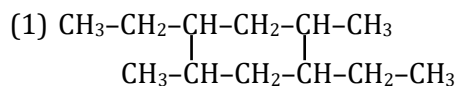
CGR106

13. The product formed in the first step of the reaction of

[JEE (Main) 2021]

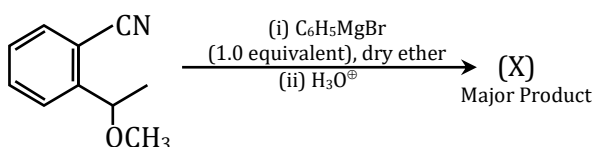


with excess $\text{Mg}/\text{Et}_2\text{O}$ ($\text{Et}=\text{C}_2\text{H}_5$) is:



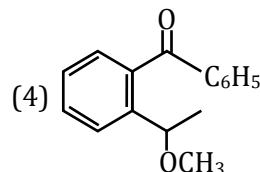
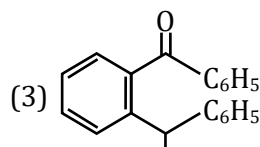
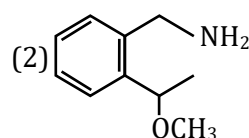
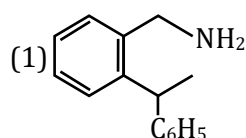
CGR035

14.



[JEE (Main) 2021]

The structure of X is :



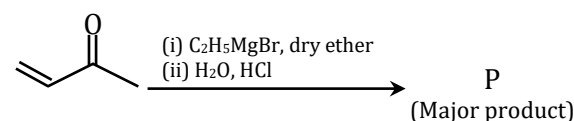
CGR036

15. To synthesise 1.0 mole of 2-methylpropan-2-ol from Ethylethanoate _____ equivalents of CH_3MgBr reagent will be required.

[JEE (Main) 2021]

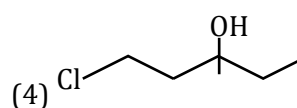
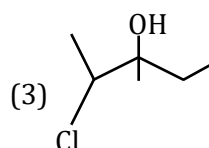
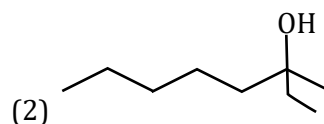
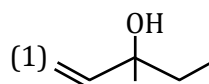
CGR037

16.



Consider the above reaction, the major product 'P' is:

[JEE (Main) 2021]



CGR095

17. The correct sequential addition of reagents in the preparation of 3-nitrobenzoic acid from benzene is:

[JEE (Main) 2021]

- (1) $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, Mg/ether , CO_2 , H_3O^+
- (2) $\text{Br}_2/\text{AlBr}_3$, NaCN , H_3O^+ , $\text{HNO}_3/\text{H}_2\text{SO}_4$
- (3) $\text{Br}_2/\text{AlBr}_3$, $\text{HNO}_3/\text{H}_2\text{SO}_4$, NaCN , H_3O^+
- (4) $\text{HNO}_3/\text{H}_2\text{SO}_4$, $\text{Br}_2/\text{AlBr}_3$, Mg/ether , CO_2 , H_3O^+

CGR096

18. Given below are two statements :

[JEE (Main) 2021]

Statement I : Ethyl pent-4-yn-oate on reaction with CH_3MgBr gives a 3°-alcohol.

Statement II : In this reaction one mole of ethyl pent-4-yn-oate utilizes two moles of CH_3MgBr .

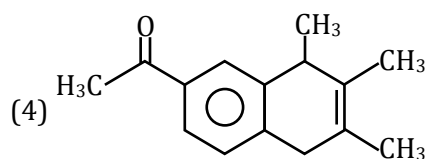
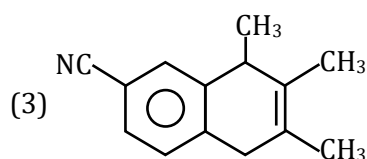
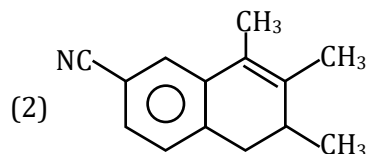
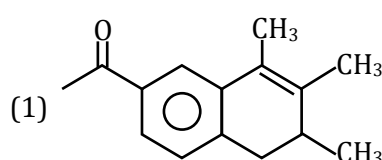
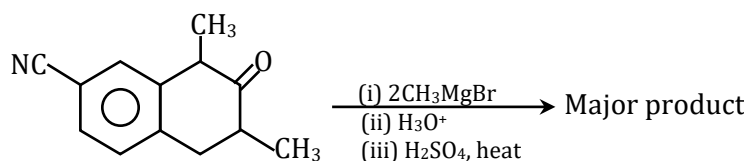
In the light of the above statements, choose the **most appropriate** answer from the options given below :

- (1) Both **Statement I** and **Statement II** are false.
- (2) **Statement I** is false but **Statement II** is true.
- (3) **Statement I** is true but **Statement II** is false.
- (4) Both **Statement I** and **Statement II** are true.

CGR039

19. Which one of the following is the major product of the given reaction?

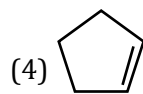
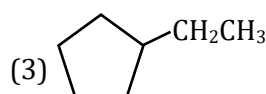
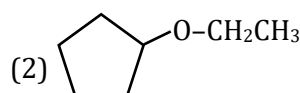
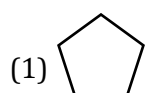
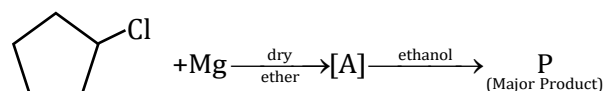
[JEE (Main) 2021]



CGR097

20. In the following sequence of reactions the P is :

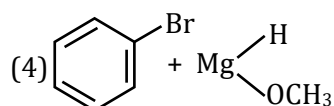
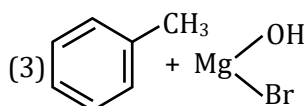
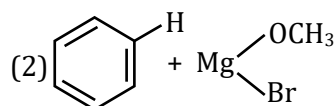
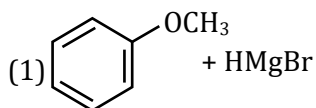
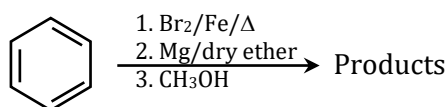
[JEE (Main) 2021]



CGR038

21. For the following :

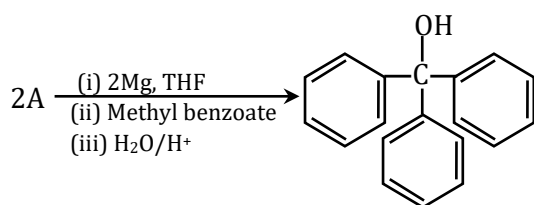
[JEE (Main) 2021]



CGR098

22. In the given reaction

[JEE (Main) 2021]



'A' can be

(1) benzyl bromide

(2) bromobenzene

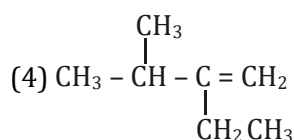
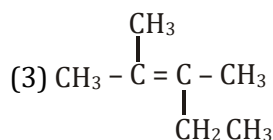
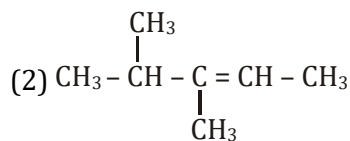
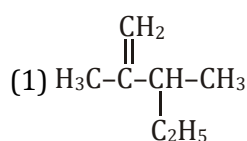
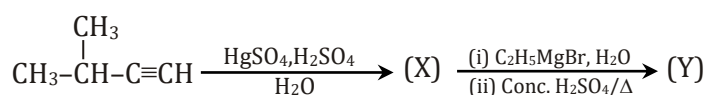
(3) cyclohexyl bromide

(4) methyl bromide

CGR099

23. The major product (Y) in the following reactions is :

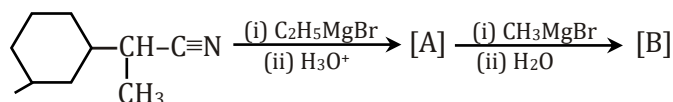
[JEE (Main) 2020]



CGR092

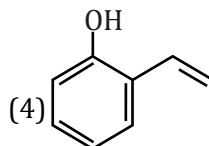
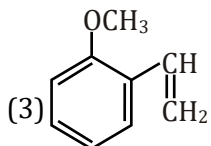
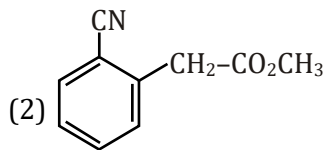
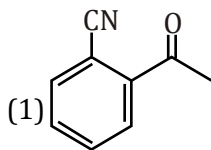
24. The number of chiral centres present in [B] is _____.

[JEE (Main) 2020]



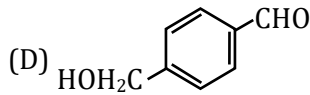
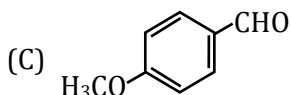
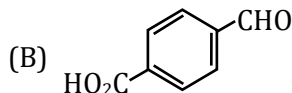
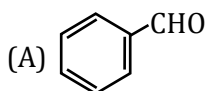
CGR093

25. Which of the following compounds reacts with ethylmagnesium bromide and also decolourizes bromine water solution :- [JEE (Main) 2019]



CGR091

26. The aldehydes which will not form Grignard product with one equivalent Grignard reagents are : [JEE (Main) 2019]



(1) (B), (C), (D)

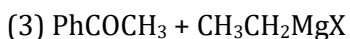
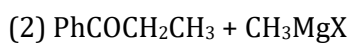
(2) (B), (D)

(3) (B), (C)

(4) (C), (D)

CGR033

27. $\text{CH}_3\text{-CH}_2\text{-}\overset{\text{OH}}{\underset{\text{Ph}}{\text{C}}}\text{-CH}_3$ cannot be prepared by :



CGR034

EXERCISE - JEE (Advanced) PYQ

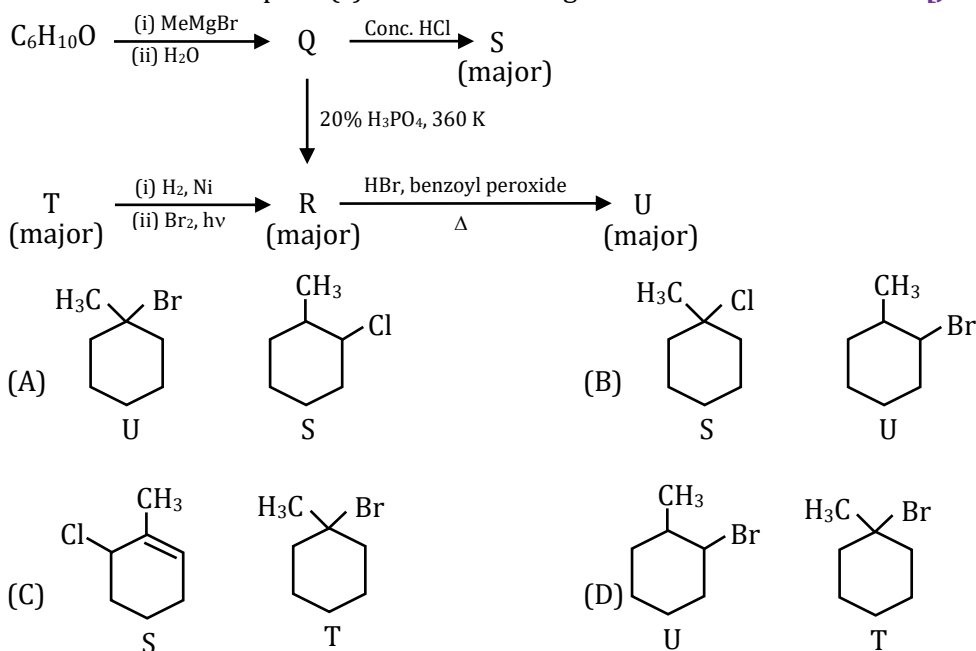
1. An organic compound **P** having molecular formula $C_6H_6O_3$ gives ferric chloride test and does not have intramolecular hydrogen bond. The compound **P** reacts with 3 equivalents of NH_2OH to produce oxime **Q**. Treatment of **P** with excess methyl iodide in the presence of KOH produces compound **R** as the major product. Reaction of **R** with excess *iso*-butylmagnesium bromide followed by treatment with H_3O^+ gives compound **S** as the major product. The total number of methyl ($-CH_3$) group(s) in compound **S** is _____.

[JEE (Advanced) 2024]

CGR131

2. Choose the correct option(s) for the following set of reactions

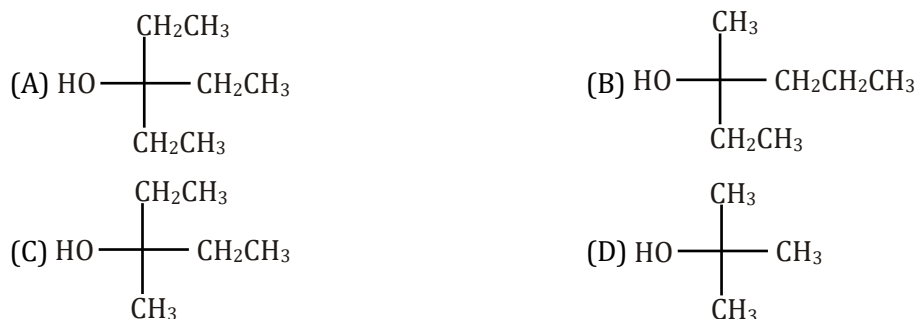
[JEE (Advanced) 2019]



CGR112

3. $CH_3MgBr + \text{Ethyl ester} \rightarrow$ which can be formed as product? (excess)

[JEE (Advanced) 2003]



CGR059

4. $(CH_3)_3CMgCl$ reaction with D_2O produces:

[JEE (Advanced) 1997]



CGR058

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	C	A	A	C	A	C	D	B
Que.	11	12	13	14						
Ans.	D	C	D	A						

EXERCISE (O-2)

Que.	1	2	3	4	5					
Ans.	A,B,C	B,C,D	A,B,C,D	B,D	B,C					

EXERCISE (S)

1.	8	2.	3	3.	0	4.	7	5.	4
6.	139	7.	3	8.	9	9.	10	10.	5
11.	6	12.	2.00	13.	6.00	14.	5.00	15.	2.00
16.	3.00								

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	1	4	1	3	2	33	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	3	4	4	2	3	4	3	1	1
Que.	21	22	23	24	25	26	27			
Ans.	2	2	3	4	4	2	1			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4						
Ans.	12,6	B,D	D	A						